

Name: _____

Date: _____



Yr 8 Mathematics (General & Ext)

Investigation, 2011

Topic – Algebra [Take Home]

Date out:

Week 6 Date 8/3/17
9

Date Due:

Week 6 Date _____

Important Information:

Although the take-home component is not worth any marks, it is essential in preparation for the in-class component. Knowledge and skills gained will be extended in the in-class validation component. This in-class validation will be completed under test conditions on the day in which this take-home component is due. The take-home component may be used when completing the in-class component.

This is an investigation called 'Opposite Corners'. You will be investigating a feature of the Hundred Square, using multiplication and subtraction. Follow the instructions and answer the questions in the spaces provided. You can draw squares and rectangles onto this Hundred Square for your investigation.

$$(2 \times 11)(1 \times 12)$$

$$22 \quad 12$$

$$\text{diff} = 10.$$

$$x \times (x+1) -$$

$$x \times (x+9) =$$

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

$$(90 \times 99) = 8910$$

$$(100 \times 89) = 8900$$

$$\underline{10}$$

1. A 2x2 square is drawn on the Hundred Square. There are two sets of opposite corners. In the square highlighted the diagonally opposite corners of the square are 46 and 57 (one set) and 47 and 56 (second set). Multiply the opposite corners and find the difference between the answers. i.e. Do 46×57 and 47×56 , then subtract the lowest answer from the highest. This is the difference.

$$2622 \quad 2632$$

$$46 \times 57 = 2622$$

$$47 \times 56 = 2632$$

$$\text{Difference} = 10$$

2. Now draw another 2x2 square onto the hundred square and repeat the process: multiply the opposite corners and find the difference between the answers. (Show your working)



3. What do you notice about your answers? Do you think this will always happen? (Explain in full sentence answers)

You always get 10.

4. Try it one more time, with a different 2x2 square to see if you are correct. (Show your working)

✓

5. Repeat the processes with a 3x3 square i.e. draw a 3x3 square onto the Hundred Square (as highlighted in the example below), multiply the opposite corners and find the difference between the answers. Show working out for three different squares.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

NB. It is only the diagonally opposite corners that you multiply. In the example above you would do 36×58 and 56×38 , then find the difference between the answers.

Example square

$$36 \times 58 = 2088$$

$$56 \times 38 = 2128$$

$$\text{Difference} = 40$$

2nd square

$$\begin{array}{r} 1 \times 23 = 23 \\ 21 \times 3 = 63 \\ \hline 40 \end{array}$$

3rd square

$$\begin{array}{r} 78 \times 100 = 7800 \\ 80 \times 98 = 7840 \\ \hline 40 \end{array}$$

6. Continue to investigate the differences between the opposite corners of squares of sizes 4x4 (Make sure you show at least three different squares in the space below.)

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

1st square

2nd square3rd square

$$\begin{array}{r}
 7 \times 40 = 280 \\
 10 \times 37 = 370 \\
 \hline
 90
 \end{array}$$

$$\begin{array}{r}
 61 \times 94 = 5734 \\
 64 \times 91 = 5824 \\
 \hline
 90
 \end{array}$$

$$\begin{array}{r}
 1 \times 34 = 34 \\
 4 \times 31 = 124 \\
 \hline
 90
 \end{array}$$

7. Continue to investigate the differences between the opposite corners of squares of sizes 5x5 (Make sure you show at least three different squares in the space below.)

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

1st square

2nd square3rd square

$$\begin{array}{r}
 1 \times 45 = 45 \\
 5 \times 41 = 205 \\
 \hline
 160
 \end{array}$$

$$\begin{array}{r}
 56 \times 100 = 5600 \\
 60 \times 96 = 5760 \\
 \hline
 160
 \end{array}$$

8. Continue to investigate the differences between the opposite corners of squares of sizes 6x6 (Make sure you show at least three different squares in the space below.)

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

1st square

2nd square3rd square

$$\begin{array}{r}
 1 \times 56 = 56 \\
 6 \times 51 = 306 \\
 \hline
 250
 \end{array}$$

9. Continue to investigate the differences between the opposite corners of squares of sizes 7x7 (Make sure you show working for at least three squares)

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

1st square

2nd square

3rd square

10. Complete the following table with your results for square size 2x2 – 7x7.

Square size (n)	Difference (D)	
2 x 2	10	$1^2 \times 10$
3 x 3	40	$2^2 \times 10$
4 x 4	90	$3^2 \times 10$
5 x 5	160	$4^2 \times 10$
6 x 6	250	$5^2 \times 10$
7 x 7	360	$6^2 \times 10$
N x N	$(n-1)^2 \times 10$	

11. Do you notice anything about your results? Are they related to a special set of numbers? Are they all multiples of the same number?

square nos.
yes, one less than the square size.

12. As you can see, there is a column in the table for you to enter difference (D), for a general square of side length (N). Write a formula below to find (D).

$$D = 10(n-1)^2$$

13. Use your rule to predict the difference in the next square size? i.e. 8 x 8 (Show working out)

1/8 Symmetrical.

$$D = 10(8-1)^2$$

$$D = 10(7)^2$$

$$D = 10 \times 49 = 490.$$